Cold Exposure / Hypothermia

Primary Assessment

If no pulse → **Start CPR** → Go to CPR CPG

Consider Spinal Motion Restriction if trauma is present

Remove wet clothing and dry patient

Determine Severity of Hypothermia

Mild
32-35°C

Cover patient with blankets (foil blanket)

Moderate & Severe
28-32°C <28 °C

Consider ALS Back-Up

Cover patient with blankets (foil blanket)

Apply heat packs to groin, axilla and neck (if available)

Consider Warmed IV Fluid
Adult: 250-500ml (max 1000ml)
Pediatric: 10ml/kg (max 20ml/kg)

EMT

Continuous ECG monitoring

Patient Care Goals

1. Rapidly identify and manage suspected cold exposure or hypothermia to prevent further heat loss and systemic complications.
2. Restore core body temperature safely and effectively to minimize secondary injury.
3. Ensure timely transport to an appropriate medical facility for advanced warming and monitoring, especially in severe cases.

Patient Inclusion & Exclusion Criteria

Inclusion Criteria

- Patients presenting with signs and symptoms suggestive of cold exposure or hypothermia, including:
  - Exposure to cold environments (e.g., wet clothing, prolonged outdoor exposure).
  - Clinical features such as apathy, ataxia, shivering, confusion, bradycardia, or diminished/absent vital signs.
  - Core temperature below 35°C.
  - History of immersion, inadequate clothing, or exhaustion in cold conditions.

Exclusion Criteria

- Hypothermia secondary to intrinsic causes such as severe sepsis or decompensated shock.

Key Patient Assessment Points

1. **Primary and Secondary Survey**
   - Check for pulse; if absent, initiate CPR as per Adult/Pediatric Cardiac Arrest CPG.
   - Consider spinal motion restriction if trauma is present.
2. **Severity Grading**
   - Mild: 32–35°C – Apathy/lethargy, ataxia, vasoconstriction, tachypnea, confusion.
   - Moderate: 28–32°C – Acute loss of consciousness, bradycardia, muscle rigidity, stupor or coma.
   - Severe: <28°C – Diminished/absent signs of life, apnea, dilated pupils, reduced or absent reflexes, dysrhythmias.
3. **Determine Triage Category**
   - Determine Triage Category as per Patient Triage Categories CPG.

Patient Management & Interventions

1. **Primary & Secondary Survey**
   - Perform Primary and Secondary Surveys as per Universal Care CPG.
1. **Consider ALS back-up**
   - Consider the requirement for ALS back-up as per Universal Care CPG.
   - ALS back-up should be routinely requested for:
2. **Determine severity**
   - Determine severity of hypothermia based on core temperature and clinical features
3. **General Management**
   - Remove patient from cold environment and prevent further heat loss.
   - Remove wet clothing and dry the patient.
4. **Supportive Care by Severity**
   - **Mild** (32–35°C):
     - Cover patient with blankets (use foil blanket if available).
   - **Moderate & Severe** (28–32°C or <28°C):
     - Cover patient with blankets (foil blanket if available).
     - **Active Rewarming**:
       - Apply heat packs to axilla and groin (if available).
       - Consider warmed IV fluid:
         - Adult: 250–500 mL (max 1000 mL).
         - Pediatric: 10 mL/kg (max 20 mL/kg).
       - Continuous ECG monitoring is required to detect arrhythmias.

        - Target temperature increase: 1–2°C/hour.

5. **Cardiac Arrest Management**
    - Active rewarming (as above).
    - Confirm Cardiac Arrest Carefully:
        - Profound hypothermia can mimic death (stiffness, fixed pupils, absent reflexes).
        - Assess for at least 60 seconds for pulse and breathing.
    - CPR & Defibrillation:
        - Begin standard CPR.
        - Defibrillation: Up to 3 shocks attempted until core temp ≥ 30 °C. Then resume standard intervals.
    - Medications:
        - Withhold medications (epinephrine and amiodarone) until core temperature is >30°C;
        - Between 30–35 °C, double the interval between doses.
    - Transport:
        - All cardiac arrest patients secondary to hypothermia must be transferred to hospital under CPR.
6. **Patient Reassessment**
    - Frequently Reassess the patient as per Universal Care CPG.

Patient Safety Considerations

- Time-Sensitive Transport
    - Patients with moderate to severe hypothermia (<32°C) or cardiac arrest should be rapidly transported to a facility with warming capabilities.
    - Communicate with Dispatch identifying hypothermia severity and estimated time of exposure.
- Avoid Harmful Interventions
    - Avoid aggressive external rewarming (e.g., hot baths) in the field, as it may cause vasodilation and worsen core temperature drop.
    - Do not massage extremities in severe cases to prevent dislodging clots.

Clinical Insights & Additional Guidance

- Fluid Warming
    - Warm fluids to 40°C if available; administer in 250 mL boluses (10 mL/kg for pediatrics).
    - If accurate fluid-warming temperature monitoring is not available, administer room-temperature fluids.
- Tympanic Thermometers
    - Use tympanic thermometers if available, recognizing their unreliability in severe cases.
    - Recognition relies heavily on clinical features due to the unreliability of tympanic thermometers.
- Medication Administration
    - For analgesia, consider reducing medication doses and extending the time between doses due to altered metabolism in hypothermia.
- Afterdrop Phenomenon
    - Monitor for a potential drop in core temperature during rewarming due to cold blood from extremities returning to the core.
- Hypothermia and Cardiac Arrest
    - Withhold epinephrine until core temperature is >30°C.
    - If medications are to be given (e.g. in prolonged arrest with partial rewarming), use standard doses but extend intervals (e.g., every 6–10 minutes) to avoid accumulation and post-rewarming catecholamine surge.
    - Defibrillation doses remain unchanged.
    - All cardiac arrest patients secondary to hypothermia must be transferred to hospital under CPR.

Documentation

- Identification: Patient's full name, age, and gender.
- Mechanism / Medical Complaint: Description of cold exposure (e.g., immersion, wet clothing), duration of exposure.

- Injuries / Information: Initial temperature (if measured), clinical features, trauma considerations.
- Signs: Vital signs, neurological status, ECG findings, severity classification.
- Treatment:
    - Interventions (e.g., blankets, heat packs, warmed IV fluids).
    - Rewarming methods and target temperature increase.
    - CPR initiation if applicable.
    - Hospital notification.
- Allergies: Known allergies to medications.
- Medications: Document any administered drugs (e.g., analgesics).
- Background: Relevant history (e.g., prior cold exposure, comorbidities like diabetes).
- Other:
    - Scene conditions and witness statements.
    - Accurate timeline of exposure and symptom onset.
    - Communication with receiving facility.

Key Performance Indicators

1. Assess and document hypothermia severity.

References

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